Solution

Dividing by x^2

$$\lim_{x\to\infty}\frac{x^3+2x+3}{(5x^2+1)} = \lim_{x\to\infty}\frac{x+\frac{2}{x}+\frac{3}{x^2}}{5+\frac{1}{x^2}} \to \infty$$

That is, $\frac{x^3+2x+3}{(5x^2+1)} \to \infty$ as $x \to \infty$.

In other words, the limit does not exist.

Note that the degree of numerator is higher than that of the denominator.

Example 9.24

Calculate $\lim_{x\to\infty}\frac{1-x^3}{3x+2}$.

Solution

Dividing by x, we get

$$\frac{1-x^3}{3x+2} = \frac{\frac{1}{x}-x^2}{3+\frac{2}{x}} \to -\infty \text{ as } x \to \infty .$$

Therefore the limit does not exist.

9.2.6 Limits of rational functions

If $R(x) = \frac{p(x)}{q(x)}$ and the degree of the polynomial $p(x)$ is greater than the degree of $q(x)$, then

$$\frac{p(x)}{q(x)} \to +\infty \text{ or } -\infty \text{ as } x \to \infty.$$

If the degree of $q(x)$ is greater than the degree of $p(x)$, then

$$\lim_{x\to\infty}\frac{p(x)}{q(x)} = 0.$$

Finally, if the degree of $p(x)$ is equal to the degree of $q(x)$, then

$$\lim_{x\to\infty}\frac{p(x)}{q(x)} = \frac{\text{coeffiecent of highest power of } x \text{ in } p(x)}{\text{coefficient of highest power of } x \text{ in } q(x)}.$$

Remark

We reemphasize that statements such as $f(x) \to \infty$ as $x \to a$, $f(x) \to -\infty$ as $x \to a$, and $f(x) \to \infty$ as $x \to \infty$, $f(x) \to -\infty$ as $x \to \infty$ mean that the limits do not exist. The symbol ∞ does not represent a number and should not be treated as a number.

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